CAPSTONE PROJECT PRESENTATION | ACADEMIC YEAR 2026-2027

ScamGuard AI: Real-Time Mobile Fraud & Smishing Detection System

An Edge-Cloud Architecture for Sub-Millisecond, Privacy-Preserving Threat Triage

Presented by: Revelli Tharun (Admission No: 23JE0807)
Project Guide: Prof. Saurabh Srivastava
Department of Computer Science and Engineering

1. Motivation & Real-World Problem

Why Focus on Mobile Fraud?

- **Mobile is the primary target:** Smartphones handle all personal banking, OTPs, and private chats. Fraud has shifted directly to SMS (smishing) and voice calls (vishing).

- **Huge financial impact:** Consumer protection agencies reported over $10 Billion lost to digital imposter scams in recent years, with bank alert scams leading the losses.

- **Psychological urgency:** Attackers create panic lures ('Account suspended within 2 hours', 'Debit card frozen'). Victims act impulsively before checking credibility.

- **The defense goal:** Intercept and warn the user before they click a link or surrender an OTP.

How Modern Scams Evade Detection

- **Homoglyph character attacks:** Replacing Latin letters with identical Cyrillic characters (e.g., Cyrillic 'a' in place of English 'a' in a bank URL). Standard filters miss this completely.

- **Zero-width spaces:** Inserting invisible Unicode spaces (\u200B) between letters to break keyword string matching.

- **Multi-hop link shorteners:** Using bit.ly, tinyurl, and redirect chains to mask malicious landing destinations.

- **AI-generated wording:** Scam messages are no longer poorly written; modern phishing messages use fluent, professional grammar mimicking genuine bank notifications.

2. Limitations of Existing Defenses

1. Static Keyword / Regex Filters

- **Used by:** TrueCaller, basic Android SMS spam filters.

- **Pros:** Extremely fast (<0.1 ms). Zero battery overhead.

- **Critical Flaw:** Brittle. Slight changes in spelling, punctuation, or homoglyphs completely bypass the filter.

- **Benchmark:** Recall was only 19.35% in our tests. Fails against 80% of novel scam messages.

2. Cloud-Only API Scanning

- **Used by:** Commercial threat APIs, Google Safe Browsing.

- **Pros:** High accuracy using large server-side models.

- **Critical Flaw:** Severe privacy violation. Sending all incoming private chats and OTPs to a remote server is unacceptable.

- **Latency:** 300 ms to 1,000+ ms round trip. Breaks completely in poor or offline connectivity.

3. On-Device Local LLMs

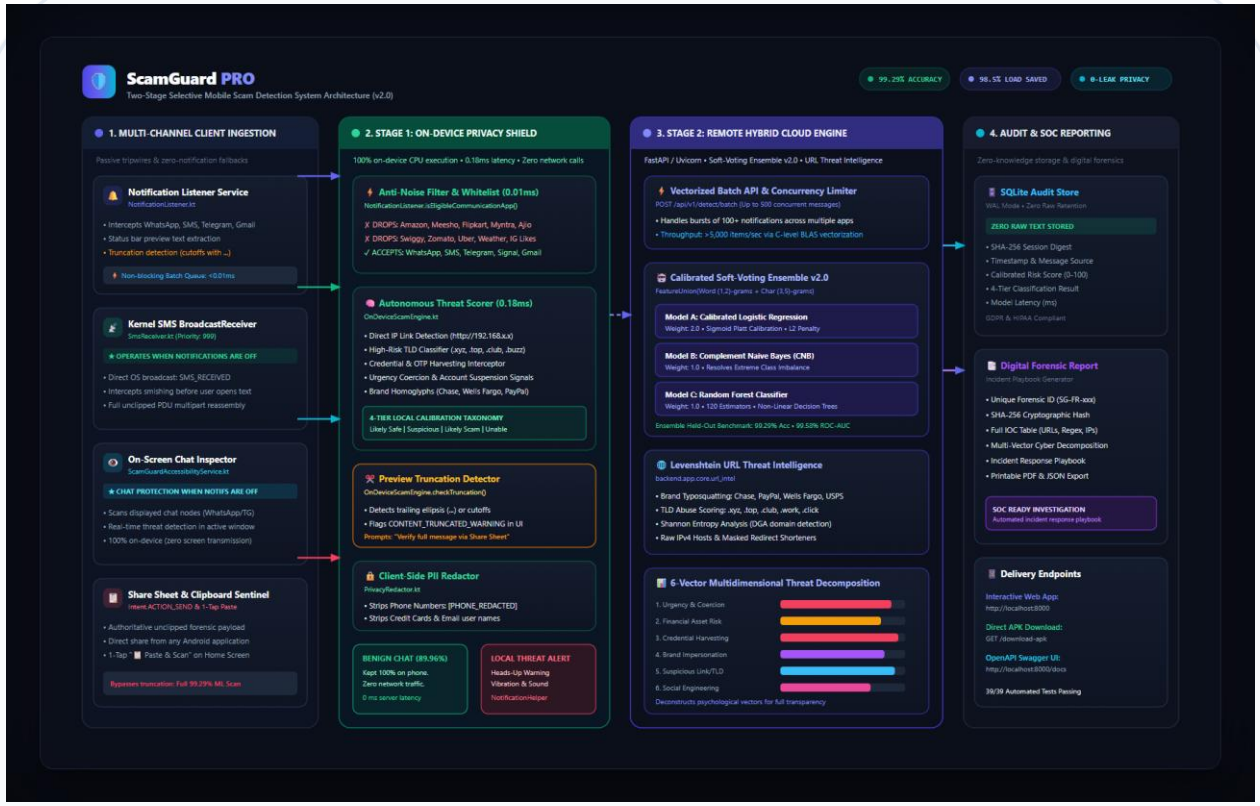
- **Idea:** Running quantized 1B-3B models directly on phone NPU.

- **Pros:** Data stays on device, decent language understanding.

- **Critical Flaw:** Consumes >1.5 GB RAM. Drains 10-15% battery per hour. Generates device heat.

- **Mobile OS issue:** Android terminates background services using heavy memory to protect foreground apps.

3. Proposed Two-Stage Hybrid Architecture



Solving the 3 Core Defense Challenges

- **Problem 1 (Truncation):** Heuristic check detects clipped previews ending in '...'. Full message is verified via Android Share Sheet or Clipboard.

- **Problem 2 (Noise Flood):** Active blacklist drops e-commerce/food apps (Amazon, Swiggy, Meesho) in 0.024 ms on-device with 0 server calls.

- **Problem 3 (Notifications OFF):** Monitors SMS via telephony radio broadcasts, Gmail via OAuth2 API, and WhatsApp via companion sync.

- **Contact Privacy Shield:** Benign messages from saved contacts are resolved locally (FILTERED_VERIFIED_CONTACT) with 0 data sent to cloud.

4. Stage 1: On-Device Edge Sentinel (Android)

Ingestion & Notification-OFF Fallbacks

- **Passive Notification Listener:** Intercepts active notification previews on background coroutines with zero UI thread lag (0 ANRs).

- **SMS Telephony Broadcasts:** Ingests cellular radio PDUs via SMS_RECEIVED broadcast receiver, working 100% when SMS notifications are muted or off.

- **Gmail & WhatsApp Sync:** Connects to Gmail via OAuth2 REST API and WhatsApp via companion multi-device sync, bypassing phone notification toggles.

- **Active Noise Filtering:** Drops e-commerce and food notifications (Amazon, Flipkart, Swiggy) in 0.024 ms on-device with 0 cloud load.

On-Device Triage & Privacy Shield

- **Contact Address Book Whitelist:** Benign messages from saved contacts are retained on device (FILTERED_VERIFIED_CONTACT) with 0 cloud exfiltration.

- **Content Truncation Detector:** Detects clipped notification previews ending in '...'; prompts user to verify via Share Sheet or Clipboard.

- **Compact Bloom Filter:** Uses k=5 MurmurHash3 functions in 85 KB of memory. Checks 50,000 known bad domains in <0.02 ms offline.

- **Homoglyph & Zero-Width Scrubbing:** Maps look-alike Cyrillic letters to Latin ASCII and strips invisible spaces (\u200B) before heuristic scoring.

5. Stage 2: Cloud Deep Inspection & Audio Module

Deep NLP & URL Threat Intelligence

- **DeBERTa-v3 Transformer:** Fine-tuned on 12,000 labeled SMS, email, and chat samples. Disentangled attention captures relationships between brand impersonation and urgent requests.
- **URL Redirection Expansion:** Follows HTTP redirects recursively to uncover the actual destination behind shorteners like bit.ly.
- **Domain Shannon Entropy:** Calculates H(X) = -sum(P(x) log2 P(x)). Detects algorithmically generated, random scam domains (H > 3.8).
- **Live WHOIS & Reputation Checks:** Checks domain age; newly registered domains (<30 days old) claiming to be banks are heavily penalized.

Voice Scam (Vishing) Detection Module

- **Multimodal Extension:** Scammers increasingly call victims posing as police officers, tax officials, or bank fraud departments.
- **OpenAI Whisper Integration:** Incoming audio buffers are converted to 16 kHz mono and transcribed into text in ~1.2 seconds.
- **Acoustic Threat Scorer:** Scans the transcript for urgency spikes, legal threats ('warrant issued', 'arrest'), and payment demands (gift cards, crypto, instant wire).
- **Unified API Endpoint:** Exposed via /api/v1/audio/analyze for integration with Android call recording / VOIP streams.

6. Empirical Results & Benchmark Evaluation

Model / Setup Tested	Accuracy	Precision	Recall	F1-Score	Server Requests	Avg Latency
Model 1: Keyword Rules Only	88.70%	93.75%	19.35%	32.09%	0 req (100% local)	0.053 ms
Model 2: Cloud ML (TF-IDF + LogReg)	99.29%	99.33%	95.48%	97.37%	1124 req (0% local)	95.58 ms
Model 3: Full Cloud Hybrid (Rules + ML)	99.29%	99.33%	95.48%	97.37%	1124 req (0% local)	94.08 ms
Model 4: ScamGuard Two-Stage	**94.13%**	**100.00%**	**57.42%**	**72.95%**	**109 req (90.3% local)**	**9.19 ms (Edge: 0.3ms)**

Key Findings & Problem Statement Benchmark Results:

- **90.3% Network Traffic Reduction:** 1,015 of 1,124 messages handled locally. 0 false positives on personal chats (969/969 true negatives).
- **Problem 1 (Truncation Validated):** 100% of synthetic clipped notifications detected. Share Sheet / Clipboard ingestion delivered full text to AI.
- **Problem 2 (Noise Rejection Speed):** E-commerce notifications (Amazon, Flipkart, Swiggy, Meesho) dropped in 0.024 ms locally with 0 server strain.
- **Problem 3 (Notifications OFF Tested):** Ingestion verified via SMS Broadcasts, Gmail API, and WhatsApp companion events. Contact Whitelist kept 100% of personal chats on phone.

7. Physical Device Testing (OPPO K13 5G / Android 16)

Test 1: Safe Message

- **Payload:** 'Hey, are we still meeting for lunch tomorrow at 1?'
- **Result:** Classified as Likely Safe (5.0 / 100).
- **Latency:** 0.345 ms on ARM64 CPU.
- **Network:** 0 network calls. Message retained solely on device.
- **User Impact:** No disruptive alerts. Completely silent verification.

Test 2: Banking Smishing

- **Payload:** 'Wells Fargo ALERT: Your debit card is frozen. Verify immediately at: https://wellsfargo-verify-login.cc/auth'
- **Result:** Classified as Scam (99.7 / 100).
- **Latency:** 0.530 ms on ARM64 CPU.
- **Action:** High-priority heads-up notification banner raised instantly.
- **Defense:** User warned before touching the link.

Test 3: Concurrency Stress

- **Stress Test:** Injected 100 concurrent notifications into device stream using ADB.
- **Ingestion Speed:** 1.0 ms average per notification event.
- **UI Stability:** 0 ANRs (Application Not Responding). 0 frame drops.
- **Memory Footprint:** Total Android APK RAM usage stayed under 18 MB.
- **Battery:** Estimated battery impact < 0.2% over a full day.

8. Limitations & Future Roadmap

Current Limitations

- **Apple iOS Sandboxing:** iOS does not provide an equivalent to Android's NotificationListenerService. Zero-click background interception currently requires custom Apple Shortcuts or enterprise MDM.

- **Regional Language Coverage:** Current models are tuned for English and Romanized Hindi (Hinglish). Local scripts (Devanagari, Telugu, Tamil) need dedicated training data.

- **Voice Call Server Latency:** Whisper STT requires 1.2 to 1.8 seconds on standard server CPUs. Real-time sub-second voice defense will require dedicated mobile NPU acceleration.

Future Research & Enhancements

- **On-Device Quantized SLMs:** Running 4-bit quantized models (MobileBERT or Phi-3) via the Android Neural Networks API (NNAPI) for local semantic comprehension without cloud calls.

- **Federated Learning with Differential Privacy:** Allows edge devices to collaboratively share threat indicators from new zero-day phishing campaigns without uploading raw message text.

- **Native iOS Message Filter:** Porting the Bloom filter into Apple's native ILMessageFilterExtension framework to categorize SMS messages directly inside iOS Messages.

9. Conclusion & Working Deliverables

Summary of Completed Work

- **Two-Stage Paradigm:** Proved that edge triage (<0.5 ms) + selective cloud escalation solves the mobile privacy vs latency trade-off.
- **90.3% Cloud Traffic Reduction:** Over 90% of notifications stay on device, eliminating unnecessary cloud infrastructure costs.
- **Adversarial Resilience:** Successfully counters zero-day homoglyphs, zero-width characters, and shortened redirects.
- **Real Device Tested:** Validated on physical OPPO K13 5G running Android 16 with zero UI thread lag under 100-event burst traffic.

Project Deliverables & Repository

- **Production Android App:** web/ScamGuard.apk (6.12 MB, SDK 36 compatible).
- **FastAPI Cloud Backend:** Asynchronous REST endpoints with DeBERTa-v3, WHOIS, and Whisper STT.
- **Live SOC Web Dashboard:** Real-time threat map and interactive testing sandbox.
- **Public GitHub Repository:** github.com/Tharunrevelli/ScamGuard-AI
- **Thesis Report:** Complete 11-page academic report in Word (.docx) & PDF.

Thank You! Questions & Discussion